

Theorem 2 (Impossibility) Proof Skeleton:
No Double-Sided Bounded Do-Intervention without a Cryptographically-Isolated Trust
Anchor

PrecedentGuard — AAAI 2027 Submission Draft

2026-06-30

Contents

1 Preliminaries and Notation	2
1.1 Causal Influence Diagram for an Agent Trajectory	2
1.2 Memory Augmentation and the Precedent Tuple	2
1.3 Bounded Do-Intervention Scheme	2
2 Threat Model	2
3 Theorem 2 (Impossibility) — Statement	3
4 Supporting Lemmas	3
4.1 Lemma 1 — Existence of Indistinguishable Intervention Pair	3
4.2 Lemma 2 — Decision-Coincidence under Indistinguishability	3
4.3 Lemma 3 — Lower Bound on the Maximum of FNR and FPR	4
4.4 Lemma 4 — Trust Anchor Restores Distinguishability	4
5 Proof of Theorem 2	4
6 Corollaries (Optional, for Section 4 of Paper)	5
7 Day-by-Day Task Checklist (14 Days)	5
8 Plan B Fallback (If Theorem 2 Fails)	6
9 Inline References (Minimal Pre-Reading)	6
10 Self-Review Checklist (Before Gate 2)	7

1 Preliminaries and Notation

1.1 Causal Influence Diagram for an Agent Trajectory

Definition 1 (Causal Influence Diagram, CID). A *Causal Influence Diagram* for an LLM agent trajectory is a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T})$ where $\mathcal{V}$ is a finite set of nodes, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of directed edges, and $\mathcal{T} : \mathcal{V} \rightarrow \{\text{INTENT}, \text{MEMORY}, \text{TOOL}, \text{OBS}, \text{DECISION}, \text{UTILITY}\}$ is a node-type assignment. We write $\text{Pa}(v)$ for the parent set of node $v \in \mathcal{V}$ and $\text{Ch}(v)$ for its child set.

[Ref: [Everitt et al. 2021, Synthese](#); [Pearl 2009 §1.2.](#)]

Remark 1. Within a single trajectory, the *decision node* $D \in \mathcal{V}$ is the guard’s binary output $D \in \{\text{ALLOW}, \text{BLOCK}\}$. The *utility node* U encodes the safety-correctness label (1 if the underlying trajectory is actually unsafe, 0 if safe).

1.2 Memory Augmentation and the Precedent Tuple

Definition 2 (Precedent Tuple). Each retrievable precedent is a five-tuple $E_i = (M_i, G_i, T_i, L_i, C_i)$, where $M_i \in \mathbb{R}^{d_M}$ is a dense semantic vector, G_i is a typed causal sub-graph, T_i is a timestamp, $L_i \in \{0, 1\}^*$ is a (possibly empty) cryptographically signed audit rule, and $C_i \in \{0, 1\}$ is a binary safety label. We partition E_i into two tiers: the *poisonable tier* $\Pi_P(E_i) := \{M_i, G_i, T_i, C_i\}$ and the *signed tier* $\Pi_S(E_i) := \{L_i\}$.

1.3 Bounded Do-Intervention Scheme

Definition 3 (Bounded Do-Intervention Scheme). A *bounded do-intervention scheme* is a tuple $\Pi = (R, \Phi, \varepsilon)$, where

- $R : \mathcal{X} \times \mathcal{B} \rightarrow \mathcal{B}^K$ is a retrieval map from query trajectory and memory bank to a top- K retrieved set;
- $\Phi : \mathcal{B}^K \times \mathbb{R}^{|D|} \rightarrow \mathbb{R}^{|D|}$ is a logit perturbation map applied to a frozen guard $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^{|D|}$;
- $\varepsilon > 0$ is a perturbation budget with the constraint $\|\Phi(\cdot, \cdot)\|_\infty \leq \varepsilon$.

The guard’s decision under Π is $D_\Pi(x) = \arg \max_k [f_\theta(x) + \Phi(R(x), f_\theta(x))]_k$.

2 Threat Model

Definition 4 (Adversary $\mathcal{A}_{p,K}$). An *intervention adversary* $\mathcal{A}_{p,K}$ is permitted to apply *do-operations* on at most K causal-parent nodes of D within a poisoning budget $p \in [0, 1]$, where p is the maximum fraction of poisonable-tier content the adversary may corrupt.

Assumption 1 (No-trust-anchor setting). The adversary has access only to the poisonable tier $\Pi_P(E_i)$ and has no signing key. The defender, in the *no-trust-anchor setting*, does not consult $\Pi_S(E_i)$ when forming Φ (equivalently, L_i does not influence the perturbation map).

[Note: This is the load-bearing condition for Theorem 2. It captures the regime where the defender lacks a verifiable, unforgeable signal.]

Definition 5 (FNR/FPR of Π). Let $\mathcal{D}$ be the trajectory distribution. The false-negative and false-positive rates of Π are

$$\text{FNR}(\Pi) = \Pr_{x \sim \mathcal{D}} [D_\Pi(x) = \text{ALLOW} \mid U(x) = 1], \quad \text{FPR}(\Pi) = \Pr_{x \sim \mathcal{D}} [D_\Pi(x) = \text{BLOCK} \mid U(x) = 0].$$

3 Theorem 2 (Impossibility) — Statement

Theorem 1 (Impossibility of Double-Sided Bounded Do-Intervention). *Let $\mathcal{G}$ be a Causal Influence Diagram satisfying Definition 1, and let $\Pi = (R, \Phi, \varepsilon)$ be any bounded do-intervention scheme under Assumption 1 (no-trust-anchor setting). For any $\delta < 1/2$, no Π can simultaneously satisfy*

$$\text{FNR}(\Pi) \leq \delta \quad \text{and} \quad \text{FPR}(\Pi) \leq \delta$$

against an adversary $\mathcal{A}_{p,K}$ with $p \cdot K \geq p_$, where p_* is a constructive threshold defined in the proof.*

Remark 2. The constant p_* depends on K , the embedding dimension d_M , and the structure of the retrieval map R . **[FILL: Provide explicit formula for p_* once the construction in Lemma 1 is filled in.]**

4 Supporting Lemmas

4.1 Lemma 1 — Existence of Indistinguishable Intervention Pair

Lemma 1 (Adversarial Indistinguishable Construction). *Under Assumption 1, there exist two trajectories x_1, x_2 and a do-intervention $\nu \in \mathcal{A}_{p,K}$ such that*

- (i) $U(x_1) = 0$ (safe ground truth) and $U(x_2) = 1$ (unsafe ground truth);
- (ii) the retrieved top- K sets satisfy $R(x_1) \stackrel{d}{=} R(\nu(x_2))$ as distributions on the retrieval space;
- (iii) the perturbation map Φ takes input distributions only, so $\Phi(R(x_1), \cdot) \stackrel{d}{=} \Phi(R(\nu(x_2)), \cdot)$.

Proof sketch. **[FILL: Construct x_2 as an unsafe trajectory whose natural memory pool $\mathcal{B}_2$ contains K items semantically distant from x_1 's natural retrieved set. Then let the adversary inject K crafted items into $\mathcal{B}_2$ such that retrieval similarity to x_1 's top- K is maximized.]**

[FILL: Argue that under the retrieval map's continuity (Lipschitz or nearest-neighbor), the resulting retrieved set is identically distributed to $R(x_1)$. Use the embedding-space Voronoi argument from [Ref: Bhagoji et al. 2018, NeurIPS] (§3.1) adapted to typed-set similarity.]

[FILL: Bound the required injection size: $K_{\text{inject}} \leq K$, which holds whenever $p \cdot |\mathcal{B}_2| \geq K$, giving the threshold $p_* = K/|\mathcal{B}_2|$.] $\square$

4.2 Lemma 2 — Decision-Coincidence under Indistinguishability

Lemma 2 (Decision Coincidence). *Let $\Pi = (R, \Phi, \varepsilon)$ be any bounded do-intervention scheme under Assumption 1. For the x_1, x_2, ν given by Lemma 1,*

$$D_\Pi(x_1) = D_\Pi(\nu(x_2)) \quad \text{with probability 1.}$$

Proof sketch. **[FILL: Since $R(x_1) \stackrel{d}{=} R(\nu(x_2))$ and Φ depends on R only (no signed L_i access by Assumption 1), $\Phi(R(x_1), \cdot) \stackrel{d}{=} \Phi(R(\nu(x_2)), \cdot)$.]**

[FILL: The frozen guard f_θ is a deterministic function of the trajectory's text representation. Argue that on x_1 and $\nu(x_2)$ the text representation of the retrieved memory chunk fed back into the guard input is identical (or has identical distribution).]

[FILL: Therefore $\arg \max_k [f_\theta + \Phi]_k$ yields the same decision.] $\square$

Remark 3. Lemma 2 is the load-bearing step. It transforms a *statistical* indistinguishability (Lemma 1) into a *decision-level* coincidence.

4.3 Lemma 3 — Lower Bound on the Maximum of FNR and FPR

Lemma 3 (Maximum-Error Lower Bound). *Under the conditions of Lemmas 1–2,*

$$\max(\text{FNR}(\Pi), \text{FPR}(\Pi)) \geq \frac{1}{2}.$$

Proof sketch. **[FILL: By Lemma 2, Π outputs the same decision on x_1 (safe) and $\nu(x_2)$ (unsafe). Without loss of generality, denote this common decision d^* .]**

[FILL: If $d^* = \text{allow}$, then on $\nu(x_2)$ we have $D_\Pi = \text{allow}$ while $U = 1$ — a false negative event.]

[FILL: If $d^* = \text{block}$, then on x_1 we have $D_\Pi = \text{block}$ while $U = 0$ — a false positive event.]

[FILL: Mix over the joint distribution of (x_1, x_2) in the adversary’s strategy: at least one of FNR or FPR exceeds $1/2$ because the constructed pair contributes equally to both rates under a balanced reduction.]

[Ref: This argument follows the Bayes-error / minimax structure of [Ref: Tsipras et al. 2019, ICLR] §3 and the indistinguishability arguments of [Ref: Cohen, Rosenfeld & Kolter 2019, ICML] §3.] $\square$

4.4 Lemma 4 — Trust Anchor Restores Distinguishability

Lemma 4 (Trust Anchor Distinguishability). *Assume the signature scheme used to produce L_i is existentially unforgeable under chosen-message attack (EUF-CMA), with security parameter λ . Then, with access to $\Pi_S(E_i) = \{L_i\}$, the indistinguishability of Lemma 1 no longer holds: there is a verifier V such that*

$$\Pr[V(R(x_1)) = 1 \wedge V(R(\nu(x_2))) = 0] \geq 1 - \text{negl}(\lambda),$$

where $\text{negl}(\lambda)$ is a function vanishing faster than any inverse polynomial.

Proof sketch. **[FILL: The genuine precedent E_1 retrieved for x_1 carries a valid L_1 signed by the trusted authority. The adversary, lacking the signing key, cannot produce a L'_2 such that $\text{Verify}(L'_2, m) = 1$ for the message m corresponding to the injected memory — except with probability $\text{negl}(\lambda)$ by EUF-CMA.]**

[FILL: Therefore the verifier V accepts $R(x_1)$ but rejects $R(\nu(x_2))$ with overwhelming probability, breaking the decision-coincidence chain of Lemma 2.]

[Ref: Standard EUF-CMA argument; see Katz & Lindell, *Introduction to Modern Cryptography* (3rd ed.), §12.4.] $\square$

5 Proof of Theorem 2

Proof of Theorem 1. The proof proceeds in four steps, instantiating the lemmas above.

Step 1 — Setup. Fix the no-trust-anchor setting of Assumption 1. Let $\Pi = (R, \Phi, \varepsilon)$ be an arbitrary bounded do-intervention scheme.

Step 2 — Construction. By Lemma 1, there exist x_1 (safe), x_2 (unsafe), and an adversarial intervention $\nu \in \mathcal{A}_{p,K}$ with $p \geq p_*$ producing the indistinguishable retrieval distributions.

Step 3 — Decision Coincidence. By Lemma 2, $D_\Pi(x_1) = D_\Pi(\nu(x_2))$ with probability 1.

Step 4 — Lower Bound. By Lemma 3, $\max(\text{FNR}, \text{FPR}) \geq 1/2$. Since this contradicts any joint bound of the form $\text{FNR} \leq \delta \wedge \text{FPR} \leq \delta$ for $\delta < 1/2$, the desired impossibility follows.

Contrast with Trust-Anchor Setting. Under access to $\Pi_S(E_i)$, Lemma 4 restores distinguishability with overwhelming probability, and the double-sided bound of Theorem 1 becomes achievable. This establishes the trust anchor as *necessary*, not merely sufficient. $\square$

6 Corollaries (Optional, for Section 4 of Paper)

Corollary 1 (Single-Sided Possibility under Stronger Assumption). *[FILL: State that under bounded statistical distance (e.g. Lipschitz retrieval), a single-sided bound (FNR only, as in SMSR) remains achievable without a trust anchor.]*

Corollary 2 (Threshold p_* Dependence). *[FILL: Express p_* as a function of K , $|\mathcal{B}|$, and embedding-space geometry. Useful for empirical p^* matching in §4.4.]*

Corollary 3 (Equivalence with the Static-Guard Limit). *[FILL: Argue that for $K = 0$ (no memory retrieval, i.e. a static guard), the impossibility reduces to the standard Bayes-error lower bound.]*

7 Day-by-Day Task Checklist (14 Days)

Week 1 — Construction and Indistinguishability

Day 1. Notation lockdown. Finalize Definitions 1–4 with PrecedentGuard v2 notation. Verify no clash with existing paper.

Day 2. Threat model formalization. Write Assumption 1 and Definition 4 in full. Identify all variables.

Day 3. Lemma 1 construction. Construct x_1, x_2, ν on a toy CID (3 memory nodes, 2 tools). Verify retrieval similarity by hand.

Day 4. Lemma 1 generalization. Lift the toy construction to the general CID. Establish $p_* = K/|\mathcal{B}_2|$.

Day 5. Lemma 1 LaTeX. Write full proof sketch with embedding-space Voronoi argument. [Ref: Bhagoji 2018 §3.1.]

Day 6. Lemma 2 setup. Argue Φ 's dependence is only on $R(\cdot)$. Show this implies Φ -output coincidence.

Day 7. Gate 1 review. Self-audit Lemmas 1–2. **Decision:** are constructions sound? If not, pivot to Plan B.

Stop-loss: if construction fails after Day 7, drop Theorem 2 and finalize Theorem 1 + corollaries only.

Week 2 — Lower Bound, Cryptographic Step, and Verification

Day 8. Lemma 3 derivation. Write the FNR/FPR lower-bound argument. Reference [Ref: Tsipras 2019 §3] and [Ref: Cohen-Rosenfeld-Kolter 2019 §3].

Day 9. Lemma 4 cryptography. Invoke EUF-CMA for Ed25519 (PrecedentGuard v2’s signature scheme). Cite Katz & Lindell §12.4.

Day 10. Theorem 2 main proof. Assemble Lemmas 1–4 into the four-step proof. Verify each link in the chain.

Day 11. Corollary draft. Write Corollaries 1–3.

Day 12. Internal review. Hand the draft to one math-strong colleague (or LLM self-review with adversarial prompt). Mark any unclear step.

Day 13. Revision. Fix all flagged steps. Tighten constants.

Day 14. Gate 2 review. Final audit. **Decision:** is the proof airtight?

Outcome A: Theorem 2 holds $\Rightarrow$ Plan A (proceed to §3 Method with double-theorem narrative).

Outcome B: Theorem 2 has fixable gap $\Rightarrow$ extend 1 week or downgrade to a weaker version (e.g. “one-sided impossibility”).

Outcome C: Theorem 2 fundamentally fails $\Rightarrow$ commit to Plan B (Theorem 1 + 3 corollaries only).

8 Plan B Fallback (If Theorem 2 Fails)

If Theorem 2 cannot be established by Day 14, the paper’s contribution structure must be revised:

- **C1** stays — CID formalization is independent of Theorem 2.
- **C2** weakens to *Theorem 1 + 3 corollaries only*.
- **C3** stays — BDIC algorithm and empirical validation are unaffected.
- **§2.2 SMSR critique** must change from “trust isolation is *necessary*” to “trust isolation enables a *stronger* guarantee”.
- **§3 Method** must motivate the trust anchor as an engineering choice rather than a theoretical necessity.
- **Target venue** may shift from AAAI 2027 to NeurIPS 2027 for an extra 3 months of polishing.

Verdict path priority: Plan A (Theorem 2 holds) $\succ$ Plan B-strong (Theorem 2-weak: only single-direction impossibility holds) $\succ$ Plan B (Theorem 2 fails entirely).

9 Inline References (Minimal Pre-Reading)

The proof draws on the following four results. **Read only the cited section, not the full paper**, unless intuition fails.

1. Madry, A. et al. (2018). *Towards Deep Learning Models Resistant to Adversarial Attacks*. ICLR. **§2 only** — adversarial budget formalization.

2. Cohen, J., Rosenfeld, E., & Kolter, J.Z. (2019). *Certified Adversarial Robustness via Randomized Smoothing*. ICML. **§3 only** — the certification-vs-attack indistinguishability structure.
3. Tsipras, D. et al. (2019). *Robustness May Be at Odds with Accuracy*. ICLR. **§3 only** — impossibility-style argument with Bayes-error bound.
4. Bhagoji, A.N., Cullina, D., & Mittal, P. (2018). *Lower Bounds on Adversarial Robustness from Optimal Transport*. NeurIPS. **§3.1 only** — embedding-space construction technique.

[Ref: [Optional deeper background:](#)] Everitt, T. et al. (2021), CID survey; Pearl, J. (2009), *Causality*, §3.4 do-operator semantics; Katz, J. & Lindell, Y. (2020), *Introduction to Modern Cryptography*, §12 EUF-CMA.

10 Self-Review Checklist (Before Gate 2)

- ☐ Definitions 1–4 are notation-compatible with PrecedentGuard v2 §2–3.
- ☐ Assumption 1 clearly excludes only the signed tier L_i .
- ☐ Lemma 1’s construction is concrete enough to compile a toy example in Python.
- ☐ Lemma 2’s argument explicitly addresses both deterministic and randomized guards.
- ☐ Lemma 3’s $1/2$ bound is matched by a constructive adversary, not just stated.
- ☐ Lemma 4’s EUF-CMA invocation is rigorous (security parameter and reduction sketch present).
- ☐ Theorem 2’s contradiction is sharp: $\max(\text{FNR}, \text{FPR}) \geq 1/2$ cannot be relaxed.
- ☐ Corollary 1 is consistent with SMSR’s published single-sided bound (sanity check).
- ☐ All three inline reference §sections have been re-read.
- ☐ One colleague has independently verified Lemmas 1–4.